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Leading end assemblies for movable partitions having an articulated leading member, and related systems and methods

Abstract

Movable partitions may include leading end assemblies comprising an articulated leading member. The articulated leading member may be coupled to a support trolley, and at least one coupling between the articulated leading member and the trolley enables the articulated leading member to move relative to the support trolley. Methods of operating a movable partition include displacing a portion of a leading end assembly to align a majority of at least one surface of the leading end assembly with a portion of an adjacent structure.

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Background/Summary

TECHNICAL FIELD

(1) Embodiments of the present disclosure are directed to leading end assemblies for movable partitions, movable partitions including leading end assemblies, and methods of operating movable partitions. In particular, embodiments of the present disclosure are directed to leading end assemblies for movable partitions comprising an articulated leading member and related assemblies, systems, and methods.

BACKGROUND

(2) Movable partitions are utilized in numerous situations and environments for a variety of purposes. Such partitions may include, for example, a movable partition including foldable or collapsible doors configured to enclose or subdivide a room or other area. Often, such partitions may be utilized simply for purposes of versatility in being able to subdivide a single large room

into multiple smaller rooms. The subdivision of a larger area may be desired, for example, to accommodate multiple groups or meetings simultaneously. In other applications, such partitions may be utilized for noise control depending, for example, on the activities taking place in a given room or portion thereof.

(3) Movable partitions may also be used to provide a security barrier, a fire barrier, or both a security barrier and a fire barrier. In such a case, the partition may be configured to automatically close upon the occurrence of a predetermined event such as the actuation of an associated alarm. For example, one or more accordion or similar folding-type partitions may be used as a security barrier, a fire barrier, or both a security barrier and a fire barrier wherein each partition is formed with a plurality of panels connected to one another with hinges. The hinged connection of the panels allows the partition to fold and collapse into a compact unit for purposes of storage when not deployed. The partition may be stored in a pocket formed in the wall of a building when in a retracted or folded state. When the partition is deployed to subdivide a single large room into multiple smaller rooms, secure an area during a fire, or for any other specified reason, the partition may be extended along an overhead track, which is often located above the movable partition in a header assembly, until the partition extends a desired distance across the room.

(4) When deployed, a leading end of the movable partition, often defined by a component known as a lead post or leading end assembly, complementarily engages another adjacent structure such as a fixed structure (e.g., a wall), a receptacle in the fixed structure, or a mating receptacle of another door. Such receptacles may be referred to as a jamb or a door post when formed in a fixed structure, or as a mating lead post when formed in another movable partition.

BRIEF SUMMARY

(5) In accordance with some embodiments of the present disclosure, movable partition systems may comprise a plurality of hingedly coupled panels movably coupled to a track, and a leading end assembly coupled to at least one panel of the plurality of hingedly coupled panels. The leading end assembly may comprise a support trolley for moving the leading end assembly along the track, and an articulated leading member coupled to the support trolley. At least one coupling between the leading member and the trolley may enable the articulated leading member to move relative to the support trolley.

(6) In accordance with additional embodiments of the present disclosure, a leading end assembly for a movable partition system may comprise a trolley for moving the leading end assembly along a track. An articulated leading member may be coupled to the trolley with an articulated coupling, a resilient coupling, and a counter-balancing coupling. The counter-balancing coupling may apply a biasing force to the articulated leading member countering a weight imbalance of the articulated leading member about the articulated coupling.

(7) In yet additional embodiments of the present disclosure, methods of operating a movable partition may comprise moving a movable partition along an overhead track between a retracted position and an extended position. The methods may further comprise contacting an adjacent structure with a portion of an articulated leading member, and responsive to contacting the adjacent structure with the articulated leading member, displacing the articulated leading member to align a majority of at least one surface of the articulated leading member with a portion of the adjacent structure.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) While the specification concludes with claims particularly pointing out and distinctly claiming which are regarded as embodiments of the present disclosure, the advantages of embodiments of the disclosure may be more readily ascertained from the description of embodiments of the

disclosure when read in conjunction with the accompanying drawings.

(2) FIG. 1 is a perspective view of a system including a movable partition in accordance with an embodiment of the present disclosure.

(3) FIG. 2 is a simplified and schematically illustrated partial cross-sectional view of the movable partition of FIG. 1.

(4) FIG. 3 is a partial cross-sectional side elevation view of a leading end assembly of the movable partition of FIG. 1.

(5) FIG. 4 is a detailed view of an articulated leading member coupled to a vertical support member of the leading end assembly of FIG. 3.

(6) FIG. 5 is a detail cross-sectional view of the articulated leading member coupled to the vertical support member of FIG. 4.

(7) FIG. 6 is a partial cross-sectional side elevation view of a leading end assembly according to an additional embodiment of the present disclosure.

(8) FIG. 7 is a simplified and schematically illustrated partial plan top view of the leading end assembly of FIG. 6.

DETAILED DESCRIPTION

(9) Illustrations presented herein are not meant to be actual views of any particular device or system, but are merely idealized representations that are employed to describe example embodiments of the present disclosure. Additionally, elements common between figures may retain the same numerical designation.

(10) In some embodiments, the leading end assemblies and movable partition systems disclosed herein may be similar to those described in U.S. patent application Ser. No. 13/169,299, filed Jun. 27, 2011 and entitled "Methods, Apparatuses, and Systems for Driving a Movable Partition with a Lead Drive Box," the disclosure of each of which is incorporated herein in its entirety by this reference.

(11) As described in above-referenced U.S. patent application Ser. No. 13/169,299, movable partitions may include a drive motor (i.e., a motor that drives the movable partition between expanded and retracted states) positioned on the movable partition proximate to the leading end assembly, for example, in a lead drive box formed by the leading end assembly.

(12) It is desirable that the lead post or leading end assembly of movable partition be substantially aligned with the fixed structure and/or mating receptacle such that the movable partition may be completely closed and an appropriate seal formed between the movable partition and the another adjacent structure. Embodiments of the present disclosure include a leading end assembly having an articulated leading member that is movably coupled (e.g., not rigidly coupled) to another portion of the leading end assembly or movable partition. For example, the leading member may be movable relative to another portion of the leading end assembly (e.g., a trolley that supports the leading end assembly from an overhead track and/or a vertical support structure) about at least one axis (e.g., without deforming the articulated leading member) while the remainder of the leading end assembly remains in the same position. Such a configuration enables the articulated leading member of the leading end assembly to adjust to one or more adjoining structures (e.g., a portion of an opposing wall, a portion of a pocket in which the movable partition is stored in a retracted state, or combinations thereof) in order to ensure that the articulated leading member is in substantially alignment with the one or more adjoining structures.

(13) Referring to FIG. 1, a movable partition system **10** is shown, including a movable partition **12** in the form of an accordion-type door. The movable partition **12** may be used as a barrier (e.g., a security barrier, a fire barrier, or both a security barrier and a fire barrier). In other embodiments, the movable partition **12** may be used, for example, to subdivide a relatively larger space into relatively smaller spaces (e.g., rooms or areas). The movable partition **12** may be formed with a plurality of panels **16** that are connected to one another with hinges or other hinge-like members **14** to form a pleated (i.e., a plicated) structure. The movable partition **12** may be suspended from an

overhead track **22** along which the movable partition **12** moves as the movable partition **12** is extended (i.e., closed) and retracted (i.e., opened). The hinged connections of the panels **16** allow the movable partition **12** to be compactly stored in a storage area such as, for example, a storage pocket **18** formed in a wall **20** of a building when in a retracted or folded state.

(14) To deploy the movable partition **12** to an extended position, the movable partition **12** is moved along the overhead track **22** to an adjoining structure positioned at an end portion of the overhead track **22**. A leading end structure of the movable partition **12** may include a leading end assembly **24** configured to engage with an adjoining structure such as, for example, an opposing wall **26**, a door jamb, or a leading end assembly of another movable partition (not shown). While the embodiment of the movable partition **12** shown and described with reference to FIG. **1** contains a single accordion-type door, additional embodiments may include multiple doors. For example, a partition may include two doors (e.g., accordion-type doors) configured to extend across a space and join together to partition a space.

(15) As discussed in further detail below, the leading end assembly **24** includes an articulated leading member **28** that is coupled to a trolley (e.g., support trolley **30** (FIG. **3**)) that supports the leading end assembly **24** from the overhead track **22**. For example, the articulated leading member **28** may be movable relative to another portion of the leading end assembly **24** (e.g., a portion of the leading end assembly **24** other than the articulated leading member **28**, such as, for example, the trolley) in two or more axes of direction relative to the remainder of the leading end assembly **24**. Such a configuration may enable the articulated leading member **28** of the leading end assembly **24** to adjust to one or more adjoining structures (e.g., a portion of the opposing wall **26**, a portion of the wall **20**, such as the pocket **18**, or combinations thereof) in order to ensure that the articulated leading member **28** is at least substantially plumb with (e.g., in substantially alignment with) the one or more adjoining structures. For example, the articulated leading member **28** of the leading end assembly **24** may substantially align by orienting at least a majority of at least a surface of the leading member **28** (e.g., orienting a majority or an entirety of the leading or trailing surface of the leading member **28**, orienting a majority or an entirety of the entire leading member **28**, or combinations thereof) with the adjoining structure.

(16) Referring to FIG. **2**, the movable partition system **10** is shown in a partial cross-sectional schematic view in a retracted state. As shown in FIG. **2**, a movable partition system **10** may include an accordion-style movable partition **12** that may include a first assembly or structure **32** and a second assembly or structure **34** that is laterally spaced from, and extends generally parallel to, the first structure **32**. Each of the two structures **32** and **34** has a trailing end **36** structurally fixed to a jamb such as, for example, a floating jamb **38** that is movable within the pocket **18**, and a leading end **40** that is attached to the leading end assembly **24**.

(17) When the movable partition **12** is in a retracted state, the movable partition **12** may be stored in the pocket **18** formed by one or more walls. For example, as shown in FIG. **2**, the pocket **18** may be provided by one or more walls extending substantially parallel to each other and extending between a floor **42** and a ceiling **44** (FIG. **1**). As mentioned above and discussed in detail below, an articulated coupling between the articulated leading member **28** of the leading end assembly **24** and another portion of the leading end assembly **24** (e.g., the support trolley **30** (FIG. **3**)) may enable at least a portion of the articulated leading member **28** (e.g., a trailing surface **46** of the articulated leading member **28**) to substantially align (e.g., be plumb with) at least a portion of the pocket **18** (e.g., an outer portion **50** of the pocket **18** formed by two wall portions flanking the opening to the pocket **18**).

(18) The articulated leading member **28** may include a leading portion such as, for example, a leading surface **48** that may be positioned adjacent to (e.g., in abutment with) an adjoining structure such as, for example, an opposing wall **26** or the leading end assembly of another movable partition (not shown). The leading surface **48** may be a distal surface of the articulated leading member **28** (e.g., the surface of an element located furthest from the point of attachment with the movable

partition **12**, the distalmost surface of the movable partition **12** located furthest from the pocket **18**). The leading end assembly **24** may further include a trailing portion such as, for example, a trailing surface **46** positioned opposite the leading surface **48**. The trailing surface **46** may be a proximal surface of the articulated leading member **28** (e.g., the surface of an element located at the point of attachment with the movable partition **12**).

(19) As mentioned above, the leading surface **48** of the leading end assembly **24** may be substantially flat (e.g., substantially planar). For example, a majority of the leading surface **48** of the leading end assembly **24** may lie in a plane oriented transverse to an intended direction of travel of the movable partition **12** along the overhead track **22** (FIG. 1).

(20) As also mentioned above and discussed in detail below, an articulated coupling between the articulated leading member **28** of the leading end assembly **24** and a portion of the leading end assembly **24** may enable at least a portion of the articulated leading member **28** (e.g., the leading surface **48** of the articulating leading member **28**) to substantially align with (e.g., be plumb with) at least a portion of the opposing wall **26** (e.g., surface **52** of the opposing wall **26** opposing the pocket **18**).

(21) As further shown in FIG. 2, the leading end assembly **24** may be sized to form a barrier at an open end of the pocket **18** of FIG. 1 (e.g., the end of the pocket **18** through which the movable partition **12** may be extended along the overhead track) when the movable partition **12** is in a retracted state. In some embodiments, the leading end assembly **24** may form a barrier substantially covering the open end of the pocket **18**. In some embodiments, the leading surface **48** of the leading end assembly **24** may be substantially flush with a portion of the pocket **18** (e.g., a portion of the wall **20** forming the pocket **18**) when the movable partition **12** is retracted in a storage position within the pocket **18**. In some embodiments, the leading surface **48** may be substantially flat and may be positioned adjacent to the opposing wall **26** in the expanded state. In other embodiments, the opposing wall **26** may include a door jamb set into the opposing wall **26** that may at least partially receive the leading end assembly **24** in the expanded state.

(22) As shown in FIG. 3 the leading end assembly **24** may include a vertical support member **60** and a diagonal support member **62** suspended from the overhead track **22** by a support trolley **30**. The vertical support member **60**, the diagonal support member **62**, and the support trolley **30** may be rigidly coupled to form a rigid triangular support structure. The leading end assembly **24** may include an articulated leading member **28** resiliently and plially coupled to the vertical support member **60**. The articulated leading member **28** may comprise the lead surface **48**, which may be rigid and substantially planar.

(23) As shown in a detailed view of the articulated leading member **28** and the vertical support member in FIG. 4. A plurality of couplings may couple the articulated leading member **28** to the vertical support member **60**, and at least one of the plurality of couplings may provide an articulated coupling between the articulated leading member **28** and the vertical support member **60**. For example, the articulated leading member **28** may be coupled to the vertical support member **60** with an articulated coupling **70**, a resilient coupling **72**, and a counter-balancing coupling **74**.

(24) The articulated coupling **70** may comprise a bolt **76** extending through a bracket **78** of the articulated leading member **28** and an aperture through the vertical support member **60**. A nut may be tightened on the bolt **76** to secure the bolt **76** to the bracket **78** of the articulated leading member **28**.

(25) As shown in a cross-sectional detail view in FIG. 5, the aperture through the vertical support member **60** may be slightly larger than an outer diameter of the bolt **76**, thus allowing the rotation and movement of the articulated leading member **28** relative to the vertical support member **60** and the support trolley **30**.

(26) The resilient coupling **72**, may comprise an annular resilient member **80** (e.g., a polyurethane die spring), positioned between a bolt **82** extending through the bracket **78** of the articulated leading member **28** and a sleeve **86** of the vertical support member **60**. For example, a segment of

metal tube may be positioned in and welded to the vertical support member **60** to provide the sleeve **86**.

(27) The counter-balancing coupling **74** may act to counter a torque about the articulated coupling **70** caused by a weight imbalance of the articulated leading member **28** about the articulated coupling **70**. As a majority of the mass of the articulated leading member **28** may be positioned on one side (e.g., the leading side) relative to a vertical reference plane extending laterally through the articulated coupling **70**, a torque may be applied about the articulated coupling **70** when the articulated leading member **28** hangs freely. This torque may cause the bolt **82** of the resilient coupling **72** to compress the annular resilient member **80**, which may result in the articulated leading member **28** hanging out of plumb.

(28) The counter-balancing coupling **74** may apply a force between the articulated leading member **28** and the vertical support member **60** that creates a torque opposing the torque caused by the weight imbalance. Additionally, the force applied by the counter-balancing coupling **74** may be adjusted by an installer to facilitate a desired orientation of the articulated leading member **28**, when the articulated leading member **28** is hanging freely. For example, the force applied by the counter-balancing coupling **74** may be adjusted to orient the articulated leading member **28** to align with a surface of the pocket **18**.

(29) Referring to FIG. **4**, the counter-balancing coupling **74** may comprise a bracket **88** attached to the articulated leading member **28** (e.g., bolted to the articulated leading member **28**) that extends around the vertical support member **60**, and a spring **90** located between the bracket **88** and the vertical support member **60**. Referring again to FIG. **5**, a first end of the spring **90** may be held in place with a bolt **92** secured to the vertical support member **60**. For example, the head of the bolt **92** may be sized to fit within an inner diameter of the spring **90** and restrict the lateral movement of the first end of the spring **90**.

(30) A spring adjustor **94** may be attached to the bracket **88** and contact a second end of the spring **90**. For example, the spring adjustor **94** may comprise a bolt **96** with a washer **98** welded to the head of the bolt **96**. A threaded portion of the bolt may be coupled to corresponding threads of the bracket **88**. For example, a nut **100** may be welded to the bracket **88** and the threaded portion of the bolt **96** may be engaged with the threads of the nut **100**.

(31) The second end of the spring **90** may be positioned over the head of the bolt **96** and rest on the washer **98**. The head of the bolt **96** may restrict the lateral movement of the second end of the spring **90**, and the washer **98** may restrict the longitudinal movement of the second end of the spring **90**.

(32) The bolt **96** may be rotated to adjust the distance between the washer **98** and the bracket **88**. Accordingly, the amount that the spring **90** is compressed may be adjusted by rotating the bolt **96**. The amount that the spring **90** is compressed may determine the amount of biasing force that is applied by the spring **90** to the bracket **88** (and thus the articulated leading member **28** coupled thereto) and to the vertical support member **60**. By rotating the bolt **96**, and thus adjusting the biasing force applied by the spring **90**, the resting orientation of the articulated leading member **28** may be adjusted.

(33) To facilitate the rotation of the bolt **96** while the spring **90** is located over the head of the bolt **96**, a set of nuts **102** may be locked together via compression at an end of the bolt **96** opposite the head. Friction between the locked nuts **102** and the threads of the bolt **96** may prevent the rotation of the nuts **102** relative to the bolt **96**, so that when a tool, such as a wrench, is applied to the nuts **102** the bolt **96** will rotate with the nuts **102**. Accordingly, the compressed length of the spring **90** and the resting orientation of the articulated leading member **28** may be easily adjusted by rotating the bolt **96**.

(34) In operation, the movable partition **12** may be moved along the overhead track **22** with a drive motor. The movable partition **12** may be moved from a retracted position within the pocket **18** to the extended position wherein the articulated leading member **28** of the movable partition **12** may

be moved into contact with a structure opposite the pocket **18**, such as an opposing wall. A portion of the articulated leading member **28** may contact a portion of the adjacent structure. In response to the contact, the articulated leading member **28** may be displaced relative to the support trolley **30** and the leading surface **48** of the articulated leading member **28** may align with a portion of the opposing wall where the articulated leading member **28** is intended to abut.

(35) Likewise, the movable partition **12** may be moved from the extended position to the retracted position within the pocket **18**. A portion of the articulated leading member **28** may contact a portion of the pocket **18**. In response to the contact, the articulated leading member **28** may be displaced relative to the support trolley **30** and the trailing surface **46** of the articulated leading member **28** may align with a portion of the pocket **18** in which the movable partition **12** is intended to be stored in the retracted position.

(36) Accordingly, the articulated leading member **28** may rotate about a longitudinal axis of the bolt of the articulated coupling **70** in response to a force applied to the articulated leading member **28**. Additionally, there may be provided sufficient space between the bolt **76** and the aperture of the articulated coupling **70**, and between the bracket **88** of the counter-balancing coupling **74** and the vertical support member **60**, to also allow rotation perpendicular to the longitudinal axis of the bolt **76**. Accordingly, the articulated leading member **28** may be configured to swivel in all directions about the articulated coupling **70**.

(37) When a force is applied to a portion of the articulated leading member **28**, such as by contact by an adjacent structure, the articulated leading member **28** may rotate about the bolt **76** of the articulated coupling **70**. Upon such rotation, the bolt **82** of the resilient coupling may apply a force to the annular resilient member **80**, and the annular resilient member **80** may compress in response to the force from the bolt **82**. When the force is no longer applied to the articulated leading member **28**, the annular resilient member **80** may resiliently expand to its original shape and cause the bolt **82**, and thus the articulated leading member **28**, to return to its original position.

(38) FIG. **6** shows an additional embodiment of a leading end assembly **110** comprising an articulated leading member **112**. The articulated leading member **112** may be coupled to the support trolley **114** via an articulated coupling **116** located at an end of a diagonal support member **118** extending from the support trolley **114**. The articulated coupling **116** may comprise a bolt **120** extending through an aperture in a bracket **122** that is attached to the articulated leading member **112** and extending through a corresponding aperture in the diagonal support member **118**. A nut, such as a self-locking nut, on the bolt **120** may be positioned so that it is not tight against the bracket **122** or the diagonal support member **118**, such that the articulated leading member **112** can swivel in all directions relative to the diagonal support member **118** about the bolt **120**. The articulated leading member **112** may additionally be coupled to the support trolley **114** by a resilient coupling **124**, which may comprise one or more compression springs **126**. For example, as shown in FIG. **7**, the resilient coupling **124** may comprise two compression springs **126** extending between the articulated leading member **112** and the support trolley **114**.

(39) The weight of the articulated leading member **112** may be suspended by a trolley **128** (see FIG. **6**) that is separate from and movable relative to the support trolley **114**. Accordingly, the leading end assembly **110** may not include any counter-balancing coupling.

(40) In view of the foregoing, leading end assemblies and movable partition systems in accordance with embodiments of the present disclosure may provide enhanced positioning of the leading end assembly of a movable partition system. For example, such leading end assemblies may be capable of automatically (e.g., without human intervention) align with one or more surrounding structures (e.g., one or both of a forward portion of a pocket in which the movable partition is storage and an opposing wall). In some embodiments, all four corners of one respective side of the leading member may be aligned with one or more surrounding structures. For example, all four corners of the trailing surface of the leading member may be aligned with a surface of the pocket when the movable partition is in a retracted position and/or all four corners of the leading surface of the

leading member may be aligned with a surface of the opposing wall when the movable partition is in an expanded position.

(41) Such alignment of the leading end assembly may also reduce the relative complexity of installation of a movable partition system as compared to conventional movable partition systems. For example, such a leading end assembly may require less adjustment of the leading end assembly in order to align portions of the movable partition systems (e.g., the leading member of the leading end assembly) with surrounding structures. In many applications, multiple structures surrounding the movable partition system may have differing orientations making it difficult, if not impossible, to align portions of the leading end assembly to each surrounding structure. Embodiments of the instant disclosure enable the leading end assembly to adjust (e.g., automatically adjust) to the differing surfaces of such surrounding structures requiring less installation time and providing enhanced alignment relative to such surrounding structures. Such enhanced alignment of the leading end assembly may be particularly beneficial where the movable partition is implemented as a fire and/or security barrier as discussed above.

(42) While particular embodiments of the disclosure have been shown and described, numerous variations and alternate embodiments encompassed by the present disclosure will occur to those skilled in the art. Accordingly, the disclosure is only limited in scope by the appended claims and their legal equivalents.

Claims

1. A movable partition system, comprising: a movable partition comprising: a plurality of hingedly coupled panels movably coupled to a track and suspended thereby, hanging in a first direction; and a leading end assembly coupled to at least one panel of the plurality of hingedly coupled panels, the leading end assembly comprising: a support trolley for moving the leading end assembly along the track; an articulated leading member extending in the first direction; a vertical support member coupled to the support trolley and extending from the support trolley, a length of the vertical support member extending along and adjacent to the articulated leading member in the first direction; a diagonal support member including a first end, proximal to the support trolley, coupled to the support trolley and a second end, distal to the support trolley, coupled to the vertical support member, the diagonal support member extending from the support trolley to the vertical support member forming a triangular support structure with the vertical support member and the support trolley; and at least one coupling connecting the articulated leading member and the vertical support member at a position on the length of the vertical support member offset from an end of the vertical support member coupled to the support trolley, the at least one coupling enables the articulated leading member to move relative to the vertical support member and the support trolley.
2. The movable partition system of claim 1, wherein the at least one coupling is configured to enable the articulated leading member to rotate about at least one axis relative to the vertical support member.
3. The movable partition system of claim 1, wherein the at least one coupling is configured to enable the articulated leading member to rotate about at least two axes relative to the vertical support member, and wherein a first axis of direction of the at least two axes of direction is perpendicular to a second axis of direction of the at least two axes of direction.
4. The movable partition system of claim 1, wherein the at least one coupling comprises an articulated coupling.
5. The movable partition system of claim 4, wherein the at least one coupling further comprises a resilient coupling.
6. The movable partition system of claim 5, wherein the resilient coupling comprises an annular resilient member positioned between a bolt extending through a bracket of the articulated leading member and a sleeve of the vertical support member.

7. The movable partition system of claim 6, wherein the annular resilient member is comprised of polyurethane.
8. The movable partition system of claim 5, further comprising a counter-balancing coupling positioned and configured to apply a biasing force to the articulated leading member to counter a weight imbalance of the articulated leading member.
9. The movable partition system of claim 8, wherein the counter-balancing coupling is adjustable to apply an adjustable force to the articulated leading member and adjust an orientation of the articulated leading member.
10. The movable partition system of claim 1, further comprising a counter-balancing coupling positioned and configured to apply a biasing force to the articulated leading member to counter a weight imbalance of the articulated leading member.
11. The movable partition system of claim 10, wherein the counter-balancing coupling comprises: a bracket attached to the articulated leading member, the bracket extending around the vertical support member; and a spring positioned between the bracket and the vertical support member.
12. The movable partition system of claim 10, wherein the biasing force applied by the counter-balancing coupling orients the articulated leading member to align with an opening to a storage pocket.
13. The movable partition system of claim 10, wherein the counter-balancing coupling comprises a compression spring.
14. The movable partition system of claim 1, wherein the articulated leading member of the leading end assembly is configured to move about an articulated coupling in order to abut with an adjoining structure when the movable partition is in an extended position.
15. The movable partition system of claim 1, wherein the articulated leading member comprises a substantially planar leading surface having a size and shape configured to cover a majority of an opening to a storage pocket for the movable partition.
16. The movable partition system of claim 1, wherein the weight of the articulated leading member is supported by a trolley separate from, and movable relative to, the support trolley.
17. A method of operating a movable partition, the method comprising: moving a movable partition along a track between a retracted position and an extended position, the movable partition comprising: a plurality of hingedly coupled panels movably coupled to the track and suspended thereby, hanging in a first direction; and a leading end assembly coupled to at least one panel of the plurality of hingedly coupled panels, the leading end assembly comprising: a support trolley for moving the leading end assembly along the track; an articulated leading member extending in the first direction; a vertical support member coupled to the support trolley and extending from the support trolley, a length of the vertical support member extending along and adjacent to the articulated leading member in the first direction; a diagonal support member including a first end, proximal to the support trolley, coupled to the support trolley and a second end, distal to the support trolley, coupled to the vertical support member, the diagonal support member extending from the support trolley to the vertical support member forming a triangular support structure with the vertical support member and the support trolley; and at least one coupling connecting the leading member and the vertical support member at a position on the length of the vertical support member offset from an end of the vertical support member coupled to the support trolley, the at least one coupling enables the articulated leading member to move relative to the vertical support member and the support trolley; contacting an adjacent structure with a portion of the articulated leading member; and responsive to contacting the adjacent structure with the articulated leading member, displacing the articulated leading member to align a majority of at least one surface of the articulated leading member with a portion of the adjacent structure.
18. The method of claim 17, wherein displacing the articulated leading member to align a majority of at least one surface of the articulated leading member with a portion of the adjacent structure comprises aligning the majority of the at least one surface of the articulated leading member with a

portion of an opposing wall that the articulated leading member is intended to abut in the extended position.

19. The method of claim 17, wherein displacing the articulated leading member to align a majority of at least one surface of the articulated leading member with a portion of the adjacent structure comprises aligning the majority of at least one surface of the articulated leading member with a portion of a pocket in which the movable partition is intended to be stored in the retracted position.

20. A movable partition system, comprising: a movable partition comprising: a plurality of hingedly coupled panels movably coupled to a track; a leading end assembly coupled to at least one panel of the plurality of hingedly coupled panels, the leading end assembly comprising: a support trolley for moving the leading end assembly along the track; an articulated leading member; a vertical support member extending from the support trolley; a diagonal support member forming a triangular support structure with the vertical support member and the support trolley; and at least one coupling connecting the articulated leading member and the vertical support member, the at least one coupling enables the articulated leading member to move relative to the vertical support member and the support trolley; and a counter-balancing coupling positioned and configured to apply a biasing force to the articulated leading member to counter a weight imbalance of the articulated leading member, the counter-balancing coupling comprising: a bracket attached to the articulated leading member, the bracket extending around the vertical support member; and a spring positioned between the bracket and the vertical support member.
